ME551 Vibrations in Linear Systems

Lecture 11

Multiple Degree of Freedom Systems 3 (Eigenvalue Problem)

Chap 4. Multi-Degree-of-Freedom System

4.1 Equations of Motion
4.2 Geometric Representation of Motion. Equilibrium Points
4.3 Stability of Equilibrium Points. Liapunov Direct Method
4.4 Linearization about Equilibrium Points
4.5 Stability of Linearized Systems
4.6 Linear Conservative Natural Systems. Symmetric Eigenvalue Problem
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4.10 Response of Multi-DOF Systems to Arbitrary Excitations
i. Undamped natural systems
ii. Systems with proportional viscous damping
iii. Systems with small damping
iv. Systems with structural damping
v. Undamped gyroscopic systems
vi. General nonconservative systems
4.11 Discrete-Time Systems

Linearized system:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{C}^*} + \underset{\text{skew sym}}{\boldsymbol{G}}\right)\dot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{K}} + \underset{\text{skew sym}}{\boldsymbol{H}}\right)\boldsymbol{q}(t) = 0$$

$$\downarrow$$

Conservative natural system: $\boldsymbol{C}^* = \boldsymbol{G} = \boldsymbol{H} = 0$

$$\downarrow$$

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0$$

4.6 Linear Conservative Natural Systems. The Symmetric Eigenvalue Problem

$$\underset{n\times n}{\boldsymbol{M}}\,\underset{n\times 1}{\ddot{\boldsymbol{q}}}+\underset{n\times n}{\boldsymbol{K}}\,\underset{n\times 1}{\boldsymbol{q}}(t)=0$$

$\downarrow\Leftarrow \boldsymbol{q}(t)=\boldsymbol{u}e^{st}$: Assumption of harmonic motion

$$s^2\boldsymbol{M}\boldsymbol{u}+\boldsymbol{K}\boldsymbol{u}=0$$

$\boldsymbol{K}\boldsymbol{u}=\lambda\boldsymbol{M}\boldsymbol{u},\lambda=-s^2\left(s=\pm i\sqrt{\lambda}\right)$: Eigenvalue problem with two symmetric matrices

$$\downarrow\Leftarrow\begin{cases}\boldsymbol{M}=\boldsymbol{Q}^T\boldsymbol{Q}\left(\text{Cholesky decomposition},\boldsymbol{Q}\text{: upper triangular}\right)\\ \boldsymbol{v}=\boldsymbol{Q}\boldsymbol{u},\ \boldsymbol{u}=\boldsymbol{Q}^{-1}\boldsymbol{v}\text{: Coordinate transformation}\left(\boldsymbol{u}=\boldsymbol{M}^{-1/2}\boldsymbol{v}\text{ for diagonal }\boldsymbol{M}\right)\\ \text{Pre-multiplication by }\boldsymbol{Q}^{-T}\end{cases}$$

$$\underbrace{\boldsymbol{Q}^{-T}\boldsymbol{K}\boldsymbol{Q}^{-1}}_{=\boldsymbol{A}=\boldsymbol{A}^T:\ n\times n\ \text{symmetric}}\boldsymbol{v}=\lambda\boldsymbol{Q}^{-T}\underbrace{\boldsymbol{M}}_{=\boldsymbol{Q}^T\boldsymbol{Q}}\boldsymbol{Q}^{-1}\boldsymbol{v}=\lambda\boldsymbol{v}$$

$\boldsymbol{A}\boldsymbol{v}=\lambda\boldsymbol{v}$: Standard Eigenvalue problem with one symmetric matrix

$\downarrow\Leftarrow\left|\boldsymbol{A}-\lambda\boldsymbol{I}\right|=0$: Characteristic equation for eigenvalues λ_r

$\left(\boldsymbol{A}-\lambda_r\boldsymbol{I}\right)\boldsymbol{v}_r=0,r=1,2,\ldots,n$: Eigenvector $\boldsymbol{v}_r$ may be multiplied by a nonzero coefficient

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Important Properties of Eigenvalues and Eigenvectors

$$\boldsymbol{A}\boldsymbol{v}_r = \lambda_r \boldsymbol{v}_r$$

$$\downarrow \Leftarrow \boldsymbol{A}\text{: Real symmetric}$$

$$\lambda_r : \text{Non-negative real}, \boldsymbol{v}_r : \text{Real}$$

- Orthogonality: $\boldsymbol{v}_r^T \boldsymbol{v}_s = 0; \boldsymbol{v}_r^T \boldsymbol{A} \boldsymbol{v}_s = 0$ for $r \neq s$: Orthogonality
- Eigen matrices: $\downarrow \Leftarrow \begin{cases} \boldsymbol{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n] : \text{Eigenvector matrix} \\ \boldsymbol{\Lambda} = [\lambda_r] : \text{Eigenvalue matrix (diagonal)} \end{cases}$

$$\boldsymbol{A}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda}$$

$$\boldsymbol{V}^T \boldsymbol{A} \boldsymbol{V} = \boldsymbol{V}^T \boldsymbol{V} \boldsymbol{\Lambda} = \boldsymbol{\Lambda} \left(\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}, \boldsymbol{v}_r^T \boldsymbol{v}_s = 0 \text{ for } r \neq s \right)$$

$$\downarrow \Leftarrow \boldsymbol{v}_r^T \boldsymbol{v}_s = \delta_{rs} \text{ then } \boldsymbol{v}_r^T \boldsymbol{A} \boldsymbol{v}_s = \lambda_r \delta_{rs} : \text{Orthonormality by Normalization}$$

$$\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}, \boldsymbol{V}^T \boldsymbol{A} \boldsymbol{V} = \boldsymbol{\Lambda}$$

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Proof of realness of Eigenvalues ; orthogonality of Eigenvectors

$\boldsymbol{A}_{n\times n}$: Real symmetric $\Rightarrow$ $\boldsymbol{A}\boldsymbol{v}_r = \lambda_r \boldsymbol{v}_r,\ r = 1, 2, \ldots, n$

Complex conjugate $\Rightarrow$ $\boldsymbol{A}\bar{\boldsymbol{v}}_r = \bar{\lambda}_r \bar{\boldsymbol{v}}_r$

$$\bar{\boldsymbol{v}}_r^T \boldsymbol{A}\boldsymbol{v}_r - \boldsymbol{v}_r^T \boldsymbol{A}\bar{\boldsymbol{v}}_r = \bar{\boldsymbol{v}}_r^T \boldsymbol{A}\boldsymbol{v}_r - \left(\bar{\boldsymbol{v}}_r^T \boldsymbol{A}\boldsymbol{v}_r\right)^T = 0 = \lambda_r \bar{\boldsymbol{v}}_r^T \boldsymbol{v}_r - \bar{\lambda}_r\left(\boldsymbol{v}_r^T \bar{\boldsymbol{v}}_r\right) = \left(\lambda_r - \bar{\lambda}_r\right)\|\boldsymbol{v}_r\|^2$$

$$\downarrow \Leftarrow \|\boldsymbol{v}_r\|^2 = \bar{\boldsymbol{v}}_r^T \boldsymbol{v}_r = \sum_{i=1}^{n} \bar{v}_{ri} v_{ri} = \sum_{i=1}^{n} |v_{ri}|^2 > 0$$

$\therefore \lambda_r$ *is real*. *Q.E.D* $\Leftarrow$ $\lambda_r - \bar{\lambda}_r = 0$

$\boldsymbol{A}_{n\times n}$: Real symmetric $\Rightarrow$ $\boldsymbol{A}\boldsymbol{v}_r = \lambda_r \boldsymbol{v}_r,\ r = 1, 2, \ldots, n$

$$\boldsymbol{A}\boldsymbol{v}_s = \lambda_s \boldsymbol{v}_s,\ s = 1, 2, \ldots, n$$

$$\boldsymbol{v}_s^T \boldsymbol{A}\boldsymbol{v}_r - \boldsymbol{v}_r^T \boldsymbol{A}\boldsymbol{v}_s = \boldsymbol{v}_s^T \boldsymbol{A}\boldsymbol{v}_r - \left(\boldsymbol{v}_r^T \boldsymbol{A}\boldsymbol{v}_s\right)^T = 0 = \lambda_r \boldsymbol{v}_s^T \boldsymbol{v}_r - \lambda_s\left(\boldsymbol{v}_r^T \boldsymbol{v}_s\right) = \left(\lambda_r - \lambda_s\right)\boldsymbol{v}_s^T \boldsymbol{v}_r$$

$$\boldsymbol{v}_s^T \boldsymbol{v}_r = 0, \quad \text{for } \lambda_r \neq \lambda_s$$

$\boldsymbol{v}_r^T \boldsymbol{v}_r = 1$ $\Leftarrow$ vector normalization (scaling)

$\boldsymbol{v}_s^T \boldsymbol{v}_r = \delta_{rs},\ r, s = 1, 2, \ldots, n$ $\Rightarrow \therefore$ *Eigenvectors are orthonormal to each other*. *Q.E.D*

$$\boldsymbol{v}_s^T \boldsymbol{A}\boldsymbol{v}_r = \lambda_r \delta_{rs},\ r, s = 1, 2, \ldots, n$$

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- **Sign properties of a real symmetric matrix *A in terms of Eigenvalues***

$$f(\boldsymbol{x}) = \boldsymbol{x}^T \underset{n\times n}{\boldsymbol{A}} \underset{n\times 1}{\boldsymbol{x}} : \text{Quadratic function}$$

$$\downarrow \Leftarrow \boldsymbol{x} = \underset{n\times n}{\boldsymbol{V}} \boldsymbol{y} : \text{Coordinate transformation with eigenvector matrix } \boldsymbol{V}$$

$$= \boldsymbol{y}^T \boldsymbol{V}^T \boldsymbol{A} \boldsymbol{V} \boldsymbol{y}$$

$$\downarrow \Leftarrow \boldsymbol{V}^T \boldsymbol{A} \boldsymbol{V} = \boldsymbol{\Lambda}$$

$$= \boldsymbol{y}^T \boldsymbol{\Lambda} \boldsymbol{y} = \sum_{i=1}^{n} \lambda_i y_i^2 \Rightarrow \begin{cases} \text{Positive definite when } \lambda_i > 0 \\ \text{Positive semi-definite when } \lambda_i \geq 0 \end{cases}$$

$$\downarrow$$

$$\boldsymbol{A} : \begin{cases} \text{Positive definite when } \lambda_i > 0 \\ \text{Positive semi-definite when } \lambda_i \geq 0 \end{cases}$$

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- **Shifting of Eigenvalues:** $\lambda_r \rightarrow (\lambda_r - \mu)$

$$\boldsymbol{A}\boldsymbol{v}_r = \lambda_r \boldsymbol{v}_r$$

$\downarrow \Leftarrow$ Subtraction of $\mu \boldsymbol{I} \boldsymbol{v}_r = \mu \boldsymbol{v}_r$ from both sides

$$(\boldsymbol{A} - \mu \boldsymbol{I})\boldsymbol{v}_r = (\lambda_r - \mu)\boldsymbol{v}_r$$

$\downarrow$

$(\lambda_r - \mu)$ is an eigenvalue of the matrix $(\boldsymbol{A} - \mu \boldsymbol{I})$

$\downarrow$

Subtraction of a constant μ from the diagonal element of $\boldsymbol{A}$ results in a shift in the eigenvalues of $\boldsymbol{A}$ by the same constant without affecting eigenvectors. $\Rightarrow$ Used to accelerate the convergence of iteration process in numerical method (Sec. 6.9 QR method for diagonalization of symmetric tri-diagonal matrix)

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- **Expansion theorem**

$$\boldsymbol{x} = \sum_{i=1}^{n} c_i \boldsymbol{v}_i = \underbrace{\left[\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\right]}_{\equiv \boldsymbol{V}} \begin{Bmatrix} c_1 \\ c_2 \\ \cdot \\ c_n \end{Bmatrix} = \boldsymbol{V}\boldsymbol{c}$$: Expansion of an arbitrary vector $\boldsymbol{x}$ (ex: physical coordinate)

$\downarrow \Leftarrow$ Premultiply by $\boldsymbol{V}^T$; $\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$

$$\boldsymbol{c} = \boldsymbol{V}^T \boldsymbol{x}$$

- **Eigenvector matrix versus modal matrix**

$$\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}; \quad \boldsymbol{V}^T\boldsymbol{A}\boldsymbol{V} = \Lambda$$

$$\updownarrow \Leftarrow \begin{cases} \boldsymbol{v} = \boldsymbol{Q}\boldsymbol{u} \text{ or } \boldsymbol{V}\left(\text{eigenvector matrix}\right) = \boldsymbol{Q}\boldsymbol{U}\left(\text{modal matrix}\right) \Leftarrow \boldsymbol{u} = \boldsymbol{Q}^{-1}\boldsymbol{v}, \\ \text{where } \boldsymbol{M} = \boldsymbol{Q}^T\boldsymbol{Q}, \boldsymbol{A} = \boldsymbol{Q}^{-T}\boldsymbol{K}\boldsymbol{Q}^{-1}\left(\text{or } \boldsymbol{K} = \boldsymbol{Q}^T\boldsymbol{A}\boldsymbol{Q}\right) \end{cases}$$

$$\boldsymbol{U}^T\boldsymbol{Q}^T\boldsymbol{Q}\boldsymbol{U} = \boldsymbol{I}; \boldsymbol{U}^T\boldsymbol{Q}^T\boldsymbol{A}\boldsymbol{Q}\boldsymbol{U} = \Lambda \text{ or}$$

$$\boldsymbol{U}^T\boldsymbol{M}\boldsymbol{U} = \boldsymbol{I}; \quad \boldsymbol{U}^T\boldsymbol{K}\boldsymbol{U} = \Lambda:$$

Orthonormality of modal matrix w.r.t. mass matrix

Orthogonality of modal matrix w.r.t. stiffness matrix

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Returning to the Original Natural Vibration Problem

$$\underset{n\times n}{\boldsymbol{M}}\,\underset{n\times 1}{\ddot{\boldsymbol{q}}}+\underset{n\times n}{\boldsymbol{K}}\,\underset{n\times 1}{\boldsymbol{q}}(t)=0,\boldsymbol{q}(0)=\boldsymbol{q}_0,\dot{\boldsymbol{q}}(0)=\dot{\boldsymbol{q}}_0$$

$$\downarrow\Leftarrow \boldsymbol{q}(t)=\boldsymbol{u}e^{st}$$

$$\boldsymbol{K}\boldsymbol{u}=\lambda\boldsymbol{M}\boldsymbol{u},\lambda=-s^2=\omega^2$$

$$\downarrow\Leftarrow \boldsymbol{u}=\boldsymbol{Q}^{-1}\boldsymbol{v},\text{ where }\boldsymbol{M}=\boldsymbol{Q}^T\boldsymbol{Q}$$

$$\boldsymbol{A}\boldsymbol{v}=\lambda\boldsymbol{v}\Leftrightarrow\det\left[\boldsymbol{A}-\lambda_r\boldsymbol{I}\right]=0,\boldsymbol{A}\boldsymbol{v}_r=\lambda_r\boldsymbol{v}_r,\ r=1,2,\ldots,n$$

$$\downarrow\Leftarrow s_r=\pm j\omega_r\ ;\ \boldsymbol{u}_r=\boldsymbol{Q}^{-1}\boldsymbol{v}_r\ \text{ and superposition principle}$$

$$\boldsymbol{q}(t)=\sum_{r=1}^{n}\boldsymbol{u}_r\left(b_r e^{+j\omega_r t}+b_r^* e^{-j\omega_r t}\right)=\sum_{r=1}^{n}B_r\boldsymbol{u}_r\cos\left(\omega_r t+\phi_r\right)$$

$$\downarrow\Leftarrow B_r\ \&\ \phi_r,r=1,2,\ldots,n\text{ from }\boldsymbol{q}_0\ \&\ \dot{\boldsymbol{q}}_0$$

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Normalized Eigen-matrix, Orthogonal Transformation, and Similarity Transformation

- **Orthonormality of eigen-matrix**:

$$\boldsymbol{AV} = \boldsymbol{V\Lambda}$$

$$\downarrow \Leftarrow \text{Normalization}$$

$$\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}\left(\boldsymbol{V}\text{: 'Orthogonal matrix'}\right)$$

$$\boldsymbol{V}^T\boldsymbol{AV} = \boldsymbol{\Lambda}$$

- **Orthogonal transformation vs similarity transformation:**

$$\boldsymbol{P}^{-1}\boldsymbol{AP} = \boldsymbol{B}: \text{ Similarity transformation}$$

$$\downarrow \Leftarrow \boldsymbol{P}^{-1} = \boldsymbol{P}^T \Leftarrow \text{Orthogonal matrix } \boldsymbol{P}: \boldsymbol{PP}^T = \boldsymbol{P}^T\boldsymbol{P} = \boldsymbol{I}$$

$$\boldsymbol{P}^T\boldsymbol{AP} = \boldsymbol{B}: \text{ Orthogonal transformation}$$

Note 1. Normalized eigen-matrix of a real symmetric matrix is a matrix for orthogonal transformation to obtain diagonal eigenvalue matrix.

Note 2. Orthogonal transformations of a real symmetric matrix do not change eigenvalues!

→ Iterative orthogonal transformations in the direction of diagonalization provide eigenvalues!

Note 3. Similarity transformations (of not only symmetric but also unsymmetric matrices) do not change eigenvalues either. (For proof, see Lecture 13: Nonsymmetric Eigenvalue Problem)

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Ex. 4.6 Eigenvalue Problem of a 3-DOF System

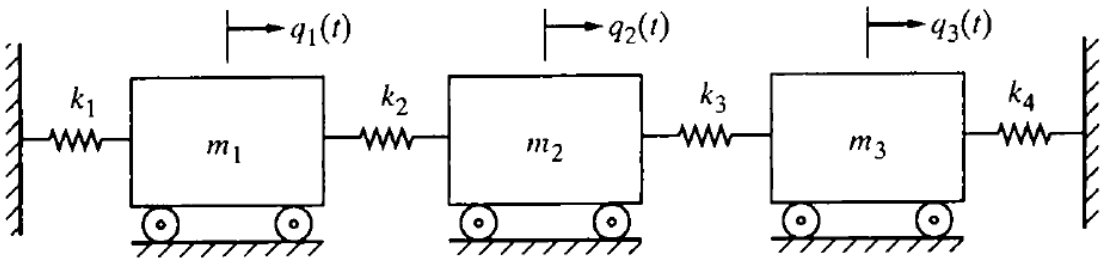

Figure 4.8 Undamped three-degree-of-freedom system

Structural parameters: $m_1 = 2m$, $m_2 = 3m$, $m_3 = m$, $k_1 = 2k$, $k_2 = 3k$, $k_3 = 2k$, $k_4 = k$

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0$$

$$T = \frac{1}{2}\left[m_1\dot{q}_1^{\,2}(t) + m_2\dot{q}_2^{\,2}(t) + m_3\dot{q}_3^{\,2}(t)\right] = \frac{1}{2}\dot{\boldsymbol{q}}^T(t)\boldsymbol{M}\dot{\boldsymbol{q}}(t)$$

$$\boldsymbol{M} = \begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} = \begin{bmatrix} 2m & 0 & 0 \\ 0 & 3m & 0 \\ 0 & 0 & m \end{bmatrix}$$

$$V = \frac{1}{2}\left\{k_1 q_1^{\,2}(t) + k_2\left[q_2(t) - q_1(t)\right]^2 + k_3\left[q_3(t) - q_2(t)\right]^2 + k_4 q_3^{\,2}(t)\right\}$$

$$= \frac{1}{2}\left\{(k_1 + k_2)q_1^{\,2}(t) + (k_2 + k_3)q_2^{\,2}(t) + (k_3 + k_4)q_3^{\,2}(t) - 2k_2 q_1(t)q_2(t) - 2k_3 q_2(t)q_3(t)\right\}$$

$$= \frac{1}{2}\boldsymbol{q}^T(t)\boldsymbol{K}\boldsymbol{q}(t)$$

$$\boldsymbol{K} = \begin{bmatrix} k_1 + k_2 & -k_2 & 0 \\ -k_2 & k_2 + k_3 & -k_3 \\ 0 & -k_3 & k_3 + k_4 \end{bmatrix} = \begin{bmatrix} 5k & -3k & 0 \\ -3k & 5k & -2k \\ 0 & -2k & 3k \end{bmatrix}$$

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Ex. 4.6 Eigenvalue Problem of a 3-DOF System

1) With one matrix A (text):

Assume harmonic response:

$$\boldsymbol{q}(t) = \boldsymbol{u}e^{st}, \lambda = \omega^2 = -s^2$$

Eigenvalue Problem:

$$\boldsymbol{Ku} = \lambda \boldsymbol{Mu}$$

$$\begin{bmatrix} 5k & -3k & 0 \\ -3k & 5k & -2k \\ 0 & -2k & 3k \end{bmatrix} \boldsymbol{u} = \lambda \begin{bmatrix} 2m & 0 & 0 \\ 0 & 3m & 0 \\ 0 & 0 & 1m \end{bmatrix} \boldsymbol{u}, \text{ where } \lambda = \omega^2$$

Cholesky decomposition:

$$\mathbf{M} = \mathbf{Q}^T\mathbf{Q}$$

$$\mathbf{Q} = \mathbf{M}^{1/2} = \begin{bmatrix} \sqrt{2m} & 0 & 0 \\ 0 & \sqrt{3m} & 0 \\ 0 & 0 & \sqrt{m} \end{bmatrix}$$

Coordinate Transform:

$$\boldsymbol{u} = \boldsymbol{Q}^{-1}\boldsymbol{v} = \boldsymbol{M}^{-1/2}\boldsymbol{v}$$

$$= m^{-1/2} \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{v}$$

$$\mathbf{A} = \mathbf{Q}^{-T}\mathbf{K}\mathbf{Q}^{-1} = \mathbf{M}^{-1/2}\mathbf{K}\mathbf{M}^{-1/2} = \frac{k}{m} \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & -3 & 0 \\ -3 & 5 & -2 \\ 0 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \frac{k}{m} \begin{bmatrix} 5/2 & -\sqrt{3/2} & 0 \\ -\sqrt{3/2} & 5/3 & -2/\sqrt{3} \\ 0 & -2/\sqrt{3} & 3 \end{bmatrix}$$

Standard EVP:

$$\boldsymbol{Av} = \lambda \boldsymbol{v} \Rightarrow \det[\boldsymbol{A} - \lambda_r \boldsymbol{I}] = 0, \boldsymbol{Av}_r = \lambda_r \boldsymbol{v}_r, \; r = 1,2,3$$

$$\lambda_1 = 0.4256\, k/m, \lambda_2 = 2.7410\, k/m, \lambda_3 = 4.0000\, k/m$$

$$\boldsymbol{v}_1 = \begin{bmatrix} 0.4743 \\ 0.8033 \\ 0.3603 \end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix} -0.7435 \\ 0.1463 \\ 0.6525 \end{bmatrix}, \boldsymbol{v}_3 = \begin{bmatrix} 0.4714 \\ -0.5774 \\ 0.6667 \end{bmatrix}$$

$$\omega_1 = 0.6527\sqrt{\frac{k}{m}}\, rad/s, \; \omega_2 = 1.6556\sqrt{\frac{k}{m}}\, rad/s, \; \omega_3 = 2.0000\sqrt{\frac{k}{m}}\, rad/s$$

$$\boldsymbol{u}_1 = m^{-1/2}\begin{bmatrix} 0.3354 \\ 0.4638 \\ 0.3603 \end{bmatrix}, \boldsymbol{u}_2 = m^{-1/2}\begin{bmatrix} -0.5257 \\ 0.0845 \\ 0.6525 \end{bmatrix}, \boldsymbol{u}_3 = m^{-1/2}\begin{bmatrix} 0.3333 \\ -0.3333 \\ 0.6667 \end{bmatrix}$$

Cf) Matlab solution:

```
>> [V,D]=eig(A)
V = 0.4743  -0.7435  -0.4714
    0.8033   0.1463   0.5774
    0.3603   0.6525  -0.6667
D =  0.4256   0        0
     0        2.7410   0
     0        0        4.0000
```

(Note: The order of eigenvalues and eigenvectors in matlab is not consistent and depends on the type of matrices)

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Ex. 4.6 Eigenvalue Problem of a 3-DOF System

1) With one matrix A (text):

Cf) Matlab solution:

```
>> A=[5/2 -sqrt(3/2) 0; -sqrt(3/2) 5/3 -2/sqrt(3); 0 -2/sqrt(3) 3]
A =
   2.5000  -1.2247        0
  -1.2247   1.6667  -1.1547
        0  -1.1547   3.0000

>> [V,D]=eig(A)
V =
   0.4743  -0.7435  -0.4714
   0.8033   0.1463   0.5774
   0.3603   0.6525  -0.6667
D =
   0.4256        0        0
        0   2.7410        0
        0        0   4.0000

>> V'*V
ans =
   1.0000        0  -0.0000
        0   1.0000  -0.0000
  -0.0000  -0.0000   1.0000

>> V'*A*V
ans =
   0.4256   0.0000  -0.0000
   0.0000   2.7410  -0.0000
  -0.0000  -0.0000   4.0000
```

Eigenvector orthogonality:

$$\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$$

Eigenvector orthogonality with respect to A:

$$\boldsymbol{V}^T\boldsymbol{A}\boldsymbol{V} = \boldsymbol{\Lambda}$$

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Ex. 4.6 Eigenvalue Problem of a 3-DOF System

2) With two matrices (M,K):

$$\boldsymbol{Ku} = \omega^2 \boldsymbol{Mu}, \text{ where } \boldsymbol{K} = \begin{bmatrix} 5k & -3k & 0 \\ -3k & 5k & -2k \\ 0 & -2k & 3k \end{bmatrix}, \boldsymbol{M} = \begin{bmatrix} 2m & 0 & 0 \\ 0 & 3m & 0 \\ 0 & 0 & m \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 5 & -3 & 0 \\ -3 & 5 & -2 \\ 0 & -2 & 3 \end{bmatrix} \boldsymbol{u} = \lambda \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{u}, \text{ where } \lambda = \frac{m\omega^2}{k}$$

$\downarrow \Leftarrow$ By matlab: [U,D]=eig(K,M), where $\boldsymbol{U}^{\mathrm{T}}\boldsymbol{MU} = \boldsymbol{I}$

```
>> M=diag([2,3,1])
M =
   2   0   0
   0   3   0
   0   0   1
>> K=[5 -3 0;-3 5 -2;0 -2 3]
K =
   5  -3   0
  -3   5  -2
   0  -2   3
>> [U, D]=eig(K, M)
U =
  0.3354  -0.5258  -0.3333
  0.4638   0.0845   0.3333
  0.3603   0.6525  -0.6667
D =
  0.4256       0       0
       0  2.7410       0
       0       0  4.0000
>> U'*M*U
ans =
   1.0000  -0.0000  -0.0000
  -0.0000   1.0000  -0.0000
  -0.0000       0   1.0000
```

$$\boldsymbol{u} = \boldsymbol{M}^{-1/2}\boldsymbol{v} = m^{-1/2} \begin{bmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{v}$$

$$\boldsymbol{u}_1 = m^{-1/2} \begin{bmatrix} 0.3354 \\ 0.4638 \\ 0.3603 \end{bmatrix}, \ \boldsymbol{u}_2 = m^{-1/2} \begin{bmatrix} -0.5257 \\ 0.0845 \\ 0.6525 \end{bmatrix}, \ \boldsymbol{u}_3 = m^{-1/2} \begin{bmatrix} 0.3333 \\ -0.3333 \\ 0.6667 \end{bmatrix}$$

Eigenvector orthogonality with respect to M and K:

```
>> U'*K*U
ans =
   0.4256  -0.0000  -0.0000
  -0.0000   2.7410   0.0000
  -0.0000  -0.0000   4.0000
```


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Highlight 11

- You can have a symmetric eigenvalue problem from conservative natural systems.

- It's eigenvalues are real and positive for positive definite mass and stiffness matrices. It's eigenvectors are orthogonal to each other with respect to mass and stiffness matrices!

- Orthogonality of eigenvectors lead us to have decoupled equations of motion in modal domain.

- The similarity (orthogonal) transformation does not change the eigenvalues but changes eigenvectors.

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